



SCRATCH

MOTION
LABPre-Workshop
Explorer GuideCreate a
Bubble Popping Game
Using Video SensingYour
Movements.
Your Game.
Your World.

Student Name: _____



School: _____



Grade: _____

Welcome to MOTION-CONTROLLED GAMING



Have you ever played a game where **your body** becomes the controller?

In this workshop, you will learn how **Scratch** can detect your movements through a webcam and use them to control a game.

Instead of pressing keys, **YOU** become part of the game.



WHAT YOU WILL LEARN



✓ Video Sensing



✓ Motion Detection



✓ Interactive Game Design



✓ Human-Computer Interaction



✓ Creative Coding



✓ Real-World Applications of Technology



QUICK POLL

Have you ever used any of these?



Xbox Kinect



Nintendo Switch
Motion Controls



VR Headsets



Gesture-Based
Games



None of These

TECHNOLOGY DETECTIVE



Can you identify where motion sensing is used?
Draw a line to match.

TECHNOLOGY

USE



Security Camera

Detects
movement



Automatic Door

Opens when
someone
approaches



Fitness Tracker

Tracks body
activity



Video Game Camera

Detects player
movement



Smartphone
Face Unlock

Detects face
position



CHALLENGE

Can you think of one more technology that
uses motion detection?



DESIGN YOUR BUBBLE POPPING GAME

Before programmers code, they plan.



MY BUBBLE THEME

- ☐  Ocean Bubbles
- ☐  Space Orbs
- ☐  Magic Potions
- ☐  Sports Balls
- ☐  Balloons
- ☐  Emoji Bubbles
- ☐  My Own Idea _____

GAME NAME



OBJECTIVE



What happens when a bubble is popped?



WHAT SHOULD PLAYERS AVOID?



Bombs



Obstacles



Time Running Out



Wrong Bubbles



My Own Idea _____



PREDICT THE TECHNOLOGY

Think like an engineer.



CHALLENGE 1

What happens if you move your hand very quickly in front of the webcam?



CHALLENGE 2

What happens if the room is very dark?



CHALLENGE 3

Will Scratch detect movement if you stay completely still?



☐ Yes ☐ No ☐ Not Sure

CHALLENGE 4

Can motion sensing be used for something other than games? Give one example.





MOTION SENSING MISSION LOG



During the workshop, tick each task as you complete it.

☐


Turn on Webcam

☐


Add Video Sensing Extension

☐


Test Camera Detection

☐


Create Bubble Sprite

☐


Make Bubble Move

☐


Detect Hand Movement

☐


Pop a Bubble

☐


Add Sound Effect

☐


Add Score

☐


Test the Game

☐


Improve the Game

☐


Save the Project



Complete all missions and your game will come to life!





INNOVATION CHALLENGE



Choose **TWO** upgrades you would like to add.

☐


Bonus Bubble

☐


Power-Up Bubble

☐


Speed Mode

☐


Penalty Bubble

☐


Timer

☐


Level System

☐


High Score System

☐


Sound Effects

☐


Multiple Bubble Types

☐


My Own Feature _____

SKETCH YOUR UPGRADE



REAL-WORLD TECHNOLOGY EXPLORER



Motion sensing is used in many industries.

Match the technology to the industry.



TECHNOLOGY

INDUSTRY



Motion Capture



Movies



Gesture Control



Gaming



Smart Cameras



Security



Physical Therapy
Tracking



Healthcare



Interactive Displays



Education

FUTURE THINKING



How could motion sensing help people in the future?



REFLECTION & INNOVATION CERTIFICATE



TODAY I LEARNED

- ☐  Motion Sensing 
- ☐  Video Detection 
- ☐  Interactive Coding 
- ☐  Game Design 
- ☐  Problem Solving 
- ☐  Creativity 

MY FAVORITE DISCOVERY



IF I HAD MORE TIME, I WOULD ADD





FUTURE INNOVATOR PLEDGE

I promise to:

- ☒ Explore technology with curiosity
- ☒ Experiment with new ideas
- ☒ Learn from mistakes
- ☒ Use technology creatively
- ☒ Build solutions that help people



Student Signature _____



YOU CODE. YOU MOVE. YOU CREATE!



You've explored the amazing world of motion sensing and interactive games using Scratch!

Keep experimenting, stay curious,
and let your creativity lead the way.

The future is yours to create!



MOVE
your body



SENSE
the motion



CODE
your ideas



POP
the bubbles



PLAY
and improve



INNOVATE
for the future

Every Great Innovator Started Somewhere.

Keep learning. Keep building. Keep making a difference!



**BE CURIOUS.
BE CREATIVE.
BE YOU.**



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